# André Pedrosa

DevOps Engineer
asp3dr0sa@gmail.com 910592261
aspedrosa in/aspedrosa



## SUMMARY

DevOps engineer with 4+ years of experience automating, deploying and running cloud-native platforms in production, building on a software development background since 2019. At Altice Labs I worked on **BotSchool**, an AI agent platform, where I developed features in **Go** and led production releases on **Kubernetes**. Experienced in infrastructure automation (Ansible, GitHub Actions, Argo CD), observability (Prometheus, Grafana, Loki) and data pipelines (Kafka, Spark, Airflow), with open-source contributions to **Apache Superset**. I use Linux as my daily driver, which makes troubleshooting across the many environments I managed come naturally. KCNA certified.

## PROFESSIONAL EXPERIENCE

**Altice Labs** *June 2023 - November 2026*
DevOps Engineer
Aveiro

**BotSchool** – cloud-native platform for creating AI agents and embedding them into clients' web platforms

- **Led the migration** of the platform's databases from Patroni-based PostgreSQL clusters to the **CloudNativePG (CNPG) operator**, moving two of the three databases with **no data loss and no impact on clients**.
  - Wrote scripts that run the migration in defined steps, rehearsed them in local environments, then ran a **production dry run** on a dummy cluster to time each step, using the same script as the real migration with a different target. The timings showed the PVCs needed more **IOPS and throughput**; raising them made the migration fit the maintenance window.
- Developed features in **Go**: metadata filtering on **Weaviate** vector stores, embedding LLM-generated picture descriptions into uploaded files parsed with **Docling**, and **MCP authentication** (PKCE, Machine-to-Machine, On-Behalf-Of) in the agent harness.
- Packaged a self-managed edition for a client **deployed on Amazon EKS**, adding health probes, security contexts and topology constraints, and adapting components to run with **least-privilege** Kubernetes permissions.
- **Ran production releases** of new platform versions, backed by an **automated integration test suite** I built to catch regressions before each release.

**Other projects**

- Contributed to migrating the Ansible automation of a product in production at MEO from **Oracle Linux 7 to 8** and from Docker to Podman, planning the rollout around component dependencies and delivering it in phases. Parallelizing the installation **cut install time from dozens of minutes to a few minutes**.
- Helped a partner team move their product from EC2 to **Amazon EKS with AWS Fargate** for on-demand workloads, automating deployments with **Helm**.
- Built **Jenkins** pipelines for RPM and container image builds, and **led their migration to GitHub Actions** when the company moved from SVN to GitHub.

- Designed a **Kafka** ingestion pipeline loading external data into a context broker and a time-series database, deploying and operating Kafka and **Argo Workflows** on **OpenShift**.
- Introduced **Prometheus, Grafana and Loki** monitoring into a platform's architecture, and researched lakehouse solutions (e.g. Delta Lake) for large-scale ETL.
- Implemented API features in Go (Gin) and Python (Django), and built an alarm backend in **Node-RED** that triggers notifications based on sensor data.



**WINPROVIT** *August 2022 - May 2023*
DevOps Engineer
Assigned to work at **Altice Labs**
Aveiro

- Rebuilt the installation automation of a multi-component platform from scratch with **Ansible**, enabling reliable deployments across **multi-host environments** and picking up Oracle Linux, Podman and distributed file systems along the way.
- Set up metrics collection, dashboards and centralized log management with **Loki**, giving the team a single view of the platform's health across hosts.
- Built an execution environment for ML models (FitEscola) and generic ETL pipelines with **PySpark and Airflow** on HDFS, plus internal libraries that abstracted Spark tuning for the team.
- Designed **Jenkins** pipelines to build and deploy the platform's components, giving the team a self-service way to ship code to every environment.




**IEETA – University of Aveiro** *June 2019 - June 2022*
Research Grant Holder, then Information Systems Technician
EHDEN project (European Health Data & Evidence Network)

- Evaluated open-source BI tools and selected **Apache Superset** to visualize and compare clinical data across partner institutions; contributed **merged pull requests** to the upstream Superset project.
- Built **NetworkDashboards** (Django, Celery, PostgreSQL, Redis) to manage and upload clinical data, with asynchronous file uploads and SQL optimizations (materialized views, query refactoring).
- Integrated both tools into the EHDEN Portal (Django) and later expanded to development of the wider portal.
- Built **GitHub Actions** CI pipelines for Superset, NetworkDashboards and the Portal, automating tests, static analysis, Docker image builds and deployment.
- Owned deployment and maintenance of Superset, NetworkDashboards and the whole EHDEN Portal in production.

## TECHNICAL SKILLS

**Cloud & containers:** Kubernetes, OpenShift, Amazon EKS, AWS Fargate, Helm, Kustomize, CloudNativePG, Docker, Podman
**CI/CD & automation:** GitHub Actions, Argo CD, Jenkins, Ansible
**Testing:** pytest, Robot Framework, Playwright
**AI platforms:** AI agents, MCP, vector stores (Weaviate), Docling, Azure AI Foundry, Vertex AI, LiteLLM
**Observability:** Prometheus, Grafana, Loki
**Languages:** Go, Python, SQL, Unix shell; familiar with Rust
**Data & messaging:** Kafka, Spark, Airflow, Argo Workflows, PostgreSQL, Redis
**Languages spoken:** Portuguese (native), English (certified by the University of Aveiro)

## CERTIFICATIONS & TRAINING

**Kubernetes and Cloud Native Associate (KCNA)** – Linux Foundation *July 2026*
**Architecting on AWS** – 3-day training course with hands-on labs *2022*

## EDUCATION

**University of Aveiro** *2019 - 2021*
**Master's degree in Informatics Engineering**
Department of Electronics, Telecommunications and Informatics

- **Thesis:** system that automates updating clinical database metadata on a central platform, with Go agents extracting data from clinical databases and a scalable **Kafka/KSQL** pipeline delivering it to metadata platforms.

**University of Aveiro** *2016 - 2019*
**Bachelor's degree in Informatics Engineering**
Department of Electronics, Telecommunications and Informatics

## ACTIVITIES & INTERESTS

**Staff Member, Portugal "O" Meeting 2026** – orienteering event with 2,500+ participants from 35 countries. Managed entries, **automated duplicate-entry detection**, helped set courses and worked on the control points team.
**Orienteering** – Orienteering Club of Estarreja since 2023; 2024 national champion in Sprint (HB) and City Race.
**Hiking** – 36 marked short-route trails (PR) across 9 districts of Portugal, 339 km in total.
**University:** Residents' Commission (2017 - 2021); GLUA, Linux Group of the University of Aveiro (2019 - 2020).
Also enjoy cycling, running and birdwatching.